

EC 101

BASICS OF ELECTRONICS & COMMUNICATION ENGINEERING

Module – V

Vol. No.-4

“Angle Modulation”



Angle Modulation

□ Angle Modulation Definition:

- **Angle Modulation** is the process of varying the total phase angle of a carrier signal in proportion with the instantaneous value of the modulating signal while keeping the amplitude of the carrier signal constant.
- The process of Frequency Modulation (FM) and Phase Modulation (PM) are two forms of angle modulation.
- FM and PM are closely related to each other, since frequency is the rate of change of phase angle.
- If either frequency or phase is varied in a modulation system, the other will change automatically as well.
- If the frequency of the carrier signal is varied *directly* in proportion to the instantaneous value of the modulating signal, then it is frequency modulation. If it is varied *indirectly*, then it is phase modulation. Therefore, **PM is sometimes known as Indirect-FM**.



Angle Modulation *contd.*

- For angle modulation, the modulated carrier signal is mathematically represented as;

$$s(t) = A_c \cos \theta_i(t) \quad [\text{Eq.5.4.1}]$$

Where, $\theta_i(t) \rightarrow$ Instantaneous angle/phase of the
modulated sinusoidal carrier

$A_c \rightarrow$ Amplitude of the modulated sinusoidal carrier

and, $\theta_i(t) = \omega_c t + \phi(t)$

where, $\omega_c \rightarrow$ Unmodulated carrier frequency

$\phi(t) \rightarrow$ Instantaneous phase deviation which is
proportional to modulating signal $m(t)$



Angle Modulation *contd.*

- $\theta_i(t)$ increases monotonically with time and changes by 2π radians in a complete oscillation. The average frequency in Hz over an interval from ' t ' to $(t + \Delta t)$, is given by;

$$f_{\Delta t}(t) = \frac{\theta_i(t + \Delta t) - \theta_i(t)}{2\pi\Delta t}$$

Thus instantaneous frequency of the angle modulated wave is;

$$f_i(t) = \lim_{\Delta t \rightarrow 0} f_{\Delta t}(t) = \lim_{\Delta t \rightarrow 0} \left[\frac{\theta_i(t + \Delta t) - \theta_i(t)}{2\pi\Delta t} \right]$$

$$\Rightarrow f_i(t) = \frac{1}{2\pi} \cdot \frac{d\theta_i(t)}{dt} \quad [\text{Eq.5.4.2}]$$

$$\Rightarrow \theta_i(t) = 2\pi \int_0^t f_i(t) \cdot dt \quad [\text{Eq.5.4.3}]$$



Angle Modulation *contd.*

- There are different ways in which the angle $\theta_i(t)$ may be varied in some manner with the baseband signal.
- However, we shall consider only two commonly used methods i.e.
 - Phase modulation (PM)
 - Frequency modulation (FM)

■ Phase Modulation (PM):

- *It is that form of angle modulation in which the angle $\theta_i(t)$ is varied linearly with the baseband signal $m(t)$, as;*

$$\theta_i(t) = \omega_c t + \phi(t) = 2\pi f_c t + k_p m(t) \quad [\text{Eq.5.4.4}]$$

Where, $2\pi f_c t \rightarrow$ angle of the unmodulated carrier

$k_p \rightarrow$ phase-sensitivity of the modulator (radians/volt)

Considering that the angle of the unmodulated carrier is zero at $t = 0$, the phase-modulated wave $s(t)$ is given by;

$$s(t) = A_c \cos[2\pi f_c t + k_p m(t)] \quad [\text{Eq.5.4.5}]$$



Angle Modulation *contd.*

- Frequency Modulation (FM):
- *It is that form of angle modulation in which the instantaneous frequency $f_i(t)$ is varied linearly with the baseband signal $m(t)$, as;*

$$f_i(t) = f_c + k_f m(t) \quad [\text{Eq.5.4.6}]$$

Where, $f_c \rightarrow$ frequency of the unmodulated carrier

$k_f \rightarrow$ frequency-sensitivity of the modulator (Hz/volt)

$$\theta_i(t) = 2\pi \int_0^t f_i(t).dt = 2\pi f_c t + 2\pi k_f \int_0^t m(t).dt$$

Considering that the angle of the unmodulated carrier is zero at $t = 0$, the frequency-modulated wave $s(t)$ is given by;

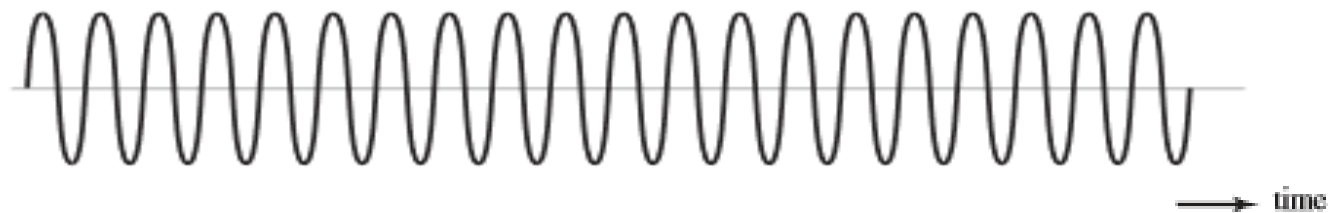
$$s(t) = A_c \cos \left[2\pi f_c t + 2\pi k_f \int_0^t m(t).dt \right] \quad [\text{Eq.5.4.7}]$$



Angle Modulation contd.

□ Properties of Angle Modulated Waves:

Carrier wave
Fig.(a)



Sinusoidal
modulating
signal
Fig.(b)



Phase modulated
signal
Fig.(c)



Frequency
modulated signal
Fig.(d)



Fig.5.4.1 [Angle modulation]



Angle Modulation *contd.*

■ **Property 1: Constancy of Transmitted Power**

From both $Eq^n(5)$ and $Eq^n(7)$ it is observed that the amplitude of PM and FM remains constant equal to the carrier amplitude A_c for all time t irrespective of the sensitivity factors k_p and k_f .

The average transmitted power of angle modulated wave is a constant, given by;

$$P_{av} = \frac{1}{2} A_c^2 \quad [\text{Eq.5.4.8}]$$

assuming load resistor of 1 ohm.

■ **Property 2: Visualization difficulty of message waveform**

In angle modulation, the message waveform can not be seen as the envelope of the modulated wave as in case of AM. Thus angle modulated wave is attributed to the **non-linear** characteristics.



Angle Modulation *contd.*

■ **Property 3: Nonlinearity of the Modulation Process**

Angle modulation is characterized by its nonlinear character, as both PM and FM waves violate the principle of superposition.

- Suppose, for example, that the message signal $m(t)$ is made up of two different components $m_1(t)$ and $m_2(t)$ as,

$$m(t) = m_1(t) + m_2(t)$$

- Let $s(t)$, $s_1(t)$ and $s_2(t)$ denotes the PM-waves produced by $m(t)$, $m_1(t)$ and $m_2(t)$ respectively. Thus,

$$s(t) = A_c \cos[2\pi f_c t + k_p(m_1(t) + m_2(t))]]$$

$$s_1(t) = A_c \cos[2\pi f_c t + k_p m_1(t)]$$

$$s_2(t) = A_c \cos[2\pi f_c t + k_p m_2(t)]$$

From the above expressions, we find that the principle of superposition is violated, because, $s(t) \neq s_1(t) + s_2(t)$



Angle Modulation *contd.*

■ **Property 4: Irregularities of Zero-crossings**

As seen from Figs. 1(c), (d), in general, the zero-crossings of a PM or FM wave no longer have a perfect regularity in their spacing across the time scale.

However, we may cite two special cases where regularity is maintained in angle modulation:

- *The message signal increases or decreases linearly with time t , in which case the instantaneous frequency of the PM wave changes from the unmodulated carrier frequency to a new constant value dependent on the slope of $m(t)$.*
- *The message signal is maintained at some constant value, positive or negative, in which case the instantaneous frequency of the FM wave changes from the unmodulated carrier frequency to a new constant value dependent on the constant value of $m(t)$.*



Angle Modulation *contd.*

- **Property 5:** *Tradeoff of increased transmission bandwidth for improved noise performance*

An important advantage of angle modulation over amplitude modulation is the realization of improved noise performance. This advantage is attributed to the fact that the transmission of a message signal by modulating the angle of a sinusoidal carrier wave is less sensitive to the presence of additive noise than transmission by modulating the amplitude of the carrier. ***The improvement in noise performance is, however, attained at the expense of a corresponding increase in the transmission bandwidth requirement of angle modulation.*** In other words, the use of angle modulation offers the possibility of exchanging an increase in transmission bandwidth for an improvement in noise performance.



Angle Modulation *contd.*

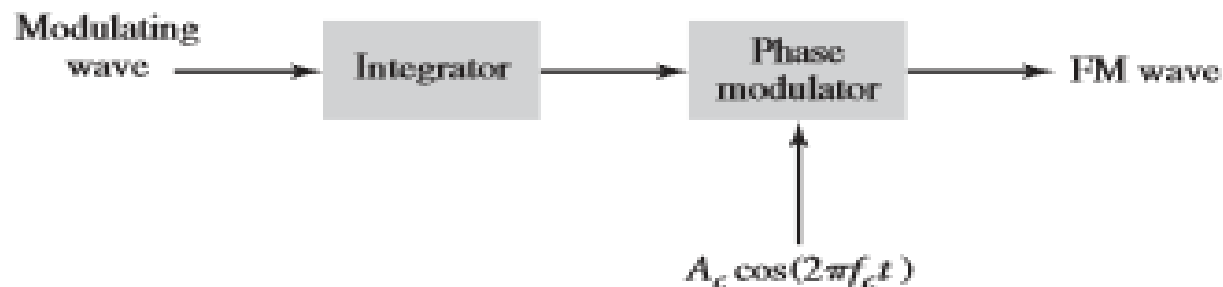
□ Relationship Between PM and FM Waves:

- Examining the definitions of E_q^n (5) and (7), we see that an FM wave may be viewed as a PM wave produced by the modulating wave $\int_0^t m(t).dt$ in place of $m(t)$. This means that an FM wave can be generated by first integrating the message signal $m(t)$ with respect to time t and then using the resulting signal as the input to a phase modulator, as shown in Fig. 2(a).
- Conversely, a PM wave can be viewed as an FM wave produced by the modulating wave $dm(t)/dt$. Hence, a PM wave can be generated by first differentiating $m(t)$ with respect to time t and then using the resulting signal as the input to a frequency modulator, as shown in Fig. 2(b).

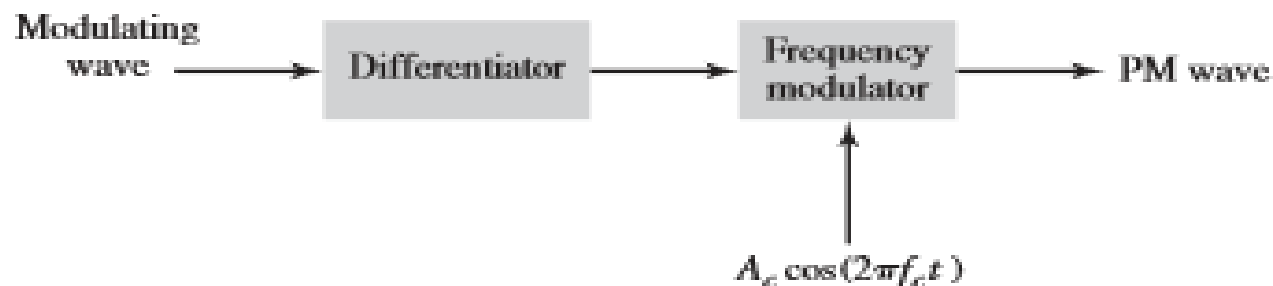
[It follows therefore that phase modulation and frequency modulation are uniquely related to each other]



Angle Modulation *contd.*



(a)



(b)

[Fig.5.4.2 [Illustration of the relationship between frequency modulation and phase modulation. (a) Scheme for generating an FM wave by using a phase modulator. (b) Scheme for generating a PM wave by using a frequency modulator.]

Single-Tone Frequency Modulation

□ Single-Tone Frequency Modulation:

- Consider a sinusoidal modulating wave defined by;

$$m(t) = A_m \cos(2\pi f_m t) \quad [\text{Eq.5.4.9}]$$

- The instantaneous frequency of the resulting FM wave equals

$$\begin{aligned} f_i(t) &= f_c + k_f A_m \cos(2\pi f_m t) \\ &= f_c + \Delta f \cos(2\pi f_m t) \end{aligned} \quad [\text{Eq.5.4.10}]$$

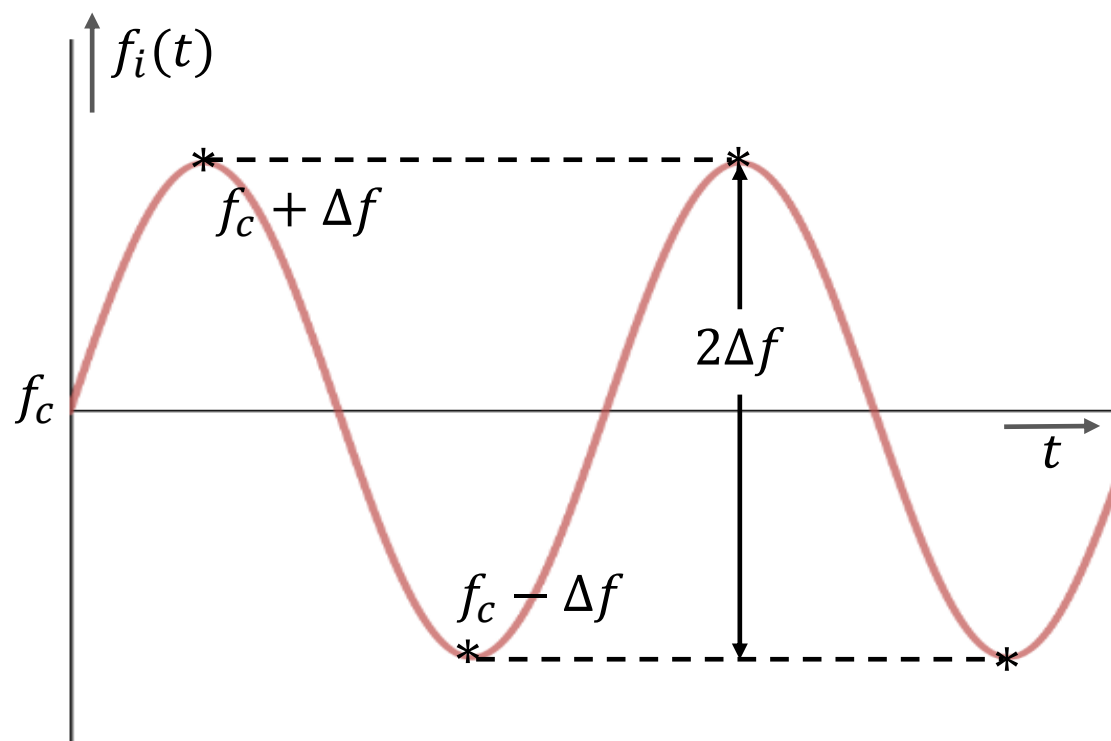
Where, $\Delta f = k_f A_m$ [Eq.5.4.11]

Frequency Deviation (Δf) is the maximum departure of the instantaneous frequency of the FM wave from the carrier frequency f_c . *Frequency deviation* (Δf) is proportional to the amplitude of the modulating wave and is independent of the modulation frequency.



Single-Tone Frequency Modulation *contd.*

- **Carrier Swing** in FM wave is defined as the peak-to-peak frequency deviation ($2\Delta f$).



$$\begin{aligned} f_i(t) \Big|_{\max} &= f_c + k_f \cdot A_m \\ &= f_c + \Delta f \end{aligned}$$

$$\begin{aligned} f_i(t) \Big|_{\min} &= f_c - k_f \cdot A_m \\ &= f_c - \Delta f \end{aligned}$$

$$\begin{aligned} \text{Peak - to - peak} \\ \text{deviation} &= 2\Delta f \end{aligned}$$

[Fig.5.4.3 [FM-wave carrier swing Vs time]



Single-Tone Frequency Modulation *contd.*

- The instantaneous angle/phase $\theta_i(t)$ of the FM wave is obtained as;

$$\begin{aligned}\theta_i(t) &= 2\pi \int_0^t f_i(t) \cdot dt \\ &= 2\pi \int_0^t [f_c + k_f m(t)] \cdot dt \\ &= 2\pi \int_0^t [f_c + k_f A_m \cos(2\pi f_m t)] \cdot dt \\ &= 2\pi f_c t + \frac{\Delta f}{f_m} \sin(2\pi f_m t) \\ &= 2\pi f_c t + \beta \sin(2\pi f_m t)\end{aligned}\quad [\text{Eq.5.4.12}]$$

Where, $\beta = \Delta f / f_m \rightarrow$ Modulation – index of FM wave.



Single-Tone Frequency Modulation *contd.*

- $\beta = \Delta f / f_m \rightarrow$ **Modulation-index** of FM wave. In a physical sense, the parameter β represents the phase deviation of the FM wave i.e. maximum departure of angle $\theta_i(t)$ from the angle $2\pi f_c t$ of the unmodulated carrier.
- Thus the *FM wave (general expression)* is given by;

$$s(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)] \quad [\text{Eq.5.4.13}]$$

- Depending on the value of the modulation-index (β), the Frequency-modulation (FM) can be classified as;
 - Narrow-band FM (NBFM) $\rightarrow \beta$ is small, $\Delta f \approx 25\text{Hz}$
 - Wide-band FM (WBFM) $\rightarrow \beta$ is large, $\Delta f \approx 75\text{KHz}$

Both β -value is compared to 1-radian.



Narrow-Band Frequency Modulation

□ Narrow-Band Frequency Modulation:

- From $Eq^n(13)$, the expression for the FM wave, for a single-tone sinusoidal modulating wave is given by;

$$\begin{aligned} s(t) &= A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)] \\ &= A_c \cos(2\pi f_c t) \cdot \cos(\beta \sin(2\pi f_m t)) \\ &\quad - A_c \sin(2\pi f_c t) \sin(\beta \sin(2\pi f_m t)) \end{aligned} \quad [Eq.5.4.14]$$

- Assuming that the modulation index is small compared to one radian, we may use the following two approximations for all times t :

$$\cos(\beta \sin(2\pi f_m t)) \simeq 1$$

$$\text{and } \sin(\beta \sin(2\pi f_m t)) \simeq \beta \sin(2\pi f_m t)$$

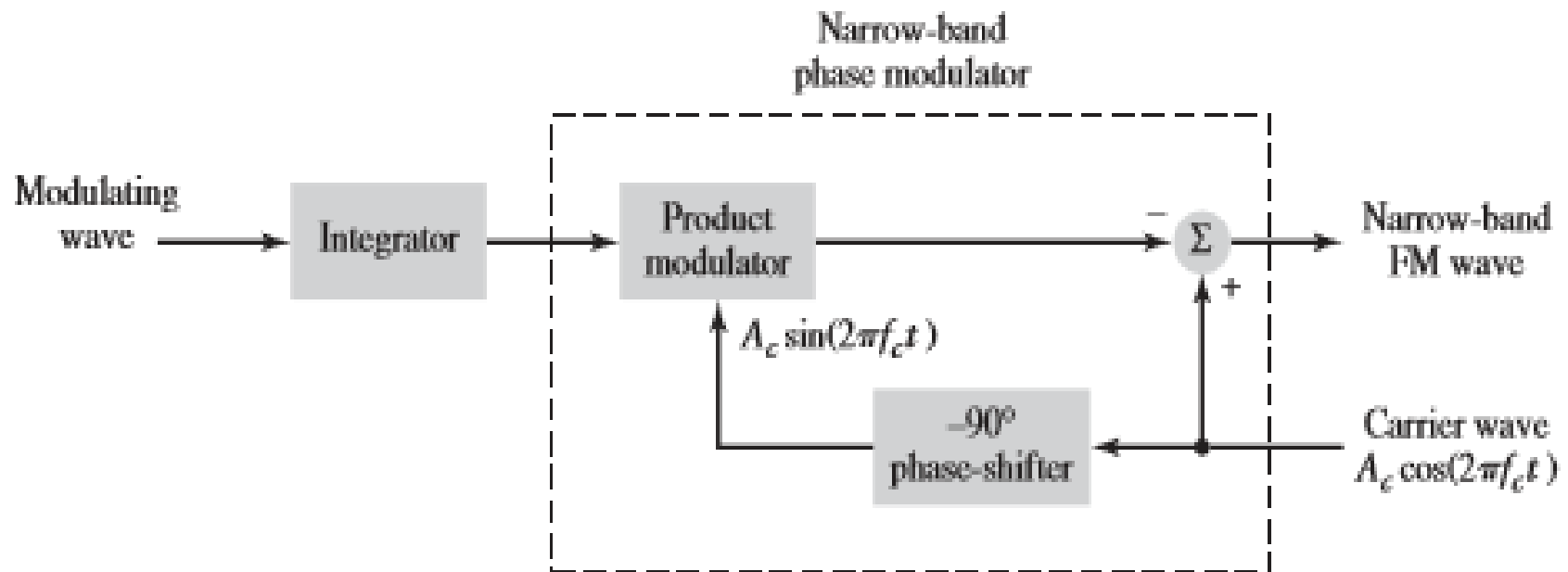
- With the above assumptions, the $Eq^n(14)$ simplifies to:

$$s(t) = A_c \cos(2\pi f_c t) - \beta A_c \sin(2\pi f_c t) \sin(2\pi f_m t) \quad [Eq.5.4.15]$$



Narrow-Band Frequency Modulation *contd.*

- [Eq.5.4.15] defines the approximate form of NBFM wave produced by a sinusoidal modulating wave $A_m \cos(2\pi f_m t)$.



[Fig.5.4.4 [Block diagram of an indirect method for generating a narrow-band FM wave]

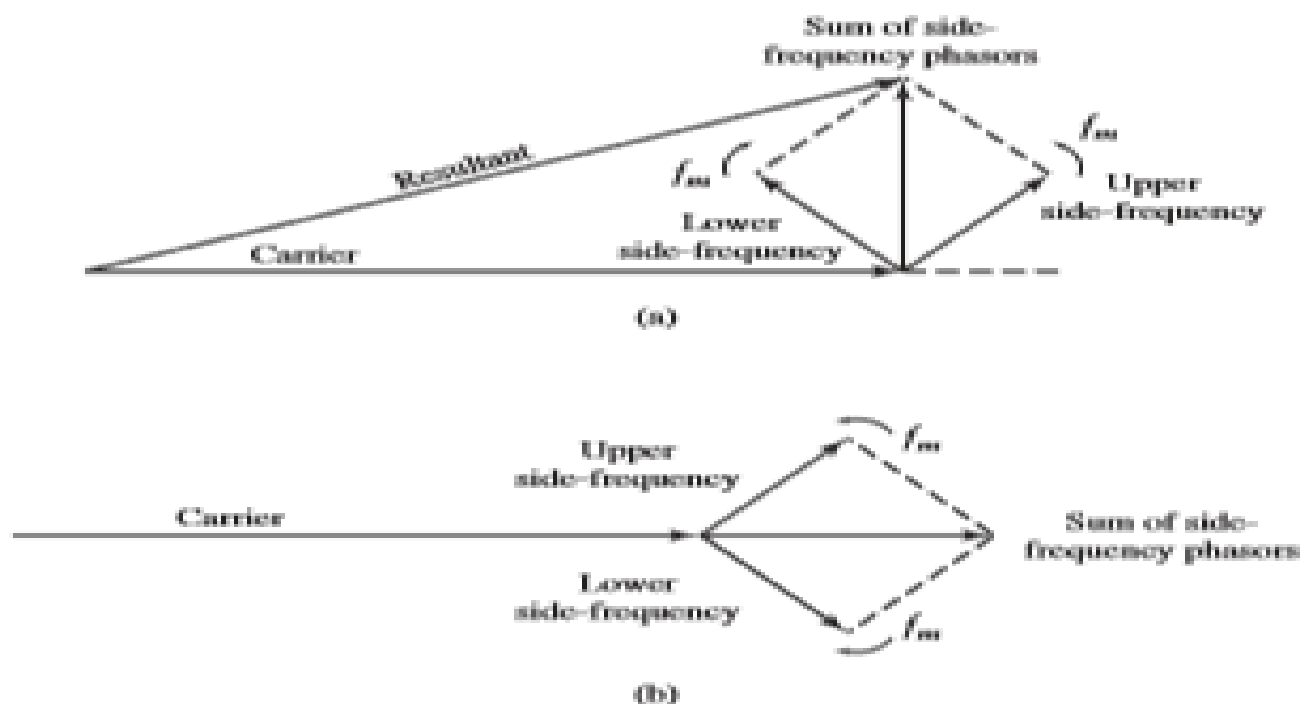
Narrow-Band Frequency Modulation contd.

- Ideally, an FM wave has a constant envelope and, for the case of a sinusoidal modulating signal of frequency f_m , the angle θ_i is also sinusoidal with the same frequency. But the modulated wave produced by the narrow-band modulator of Fig. 4 differs from this ideal condition in two fundamental respects:
 1. The envelope contains a residual amplitude modulation that varies with time.
 2. The angle θ_i contains harmonic distortion in the form of third- and higher order harmonics of the modulation frequency f_m .



Narrow-Band Frequency Modulation *contd.*

- Phasor comparison of Narrow-band FM and AM waves for sinusoidal modulation:



[Fig.5.4.5 [Phasor comparison of narrow-band FM and AM waves for sinusoidal modulation. (a) Narrow-band FM wave. (b) AM wave]

Wide-Band Frequency Modulation

□ Wide-Band Frequency Modulation (WBFM):

- There are essentially two basic methods of generating WBFM-waves, as;
 - Indirect or Armstrong Method
 - Direct Method
- In **Indirect/Armstrong method**, the message signal is first used to produce a narrow-band FM, which is followed by frequency multiplication to increase the frequency deviation to the desired level to produce the final WBFM wave.
- The **Direct method** uses a sinusoidal oscillator, with one of the reactive elements (e.g., capacitive element) in the tank circuit of the oscillator being directly controllable by the message signal, i.e. the carrier frequency is directly varied in-accordance with the input message signal.



Armstrong Method of FM Generation

■ Indirect/Armstrong Method of FM Generation:

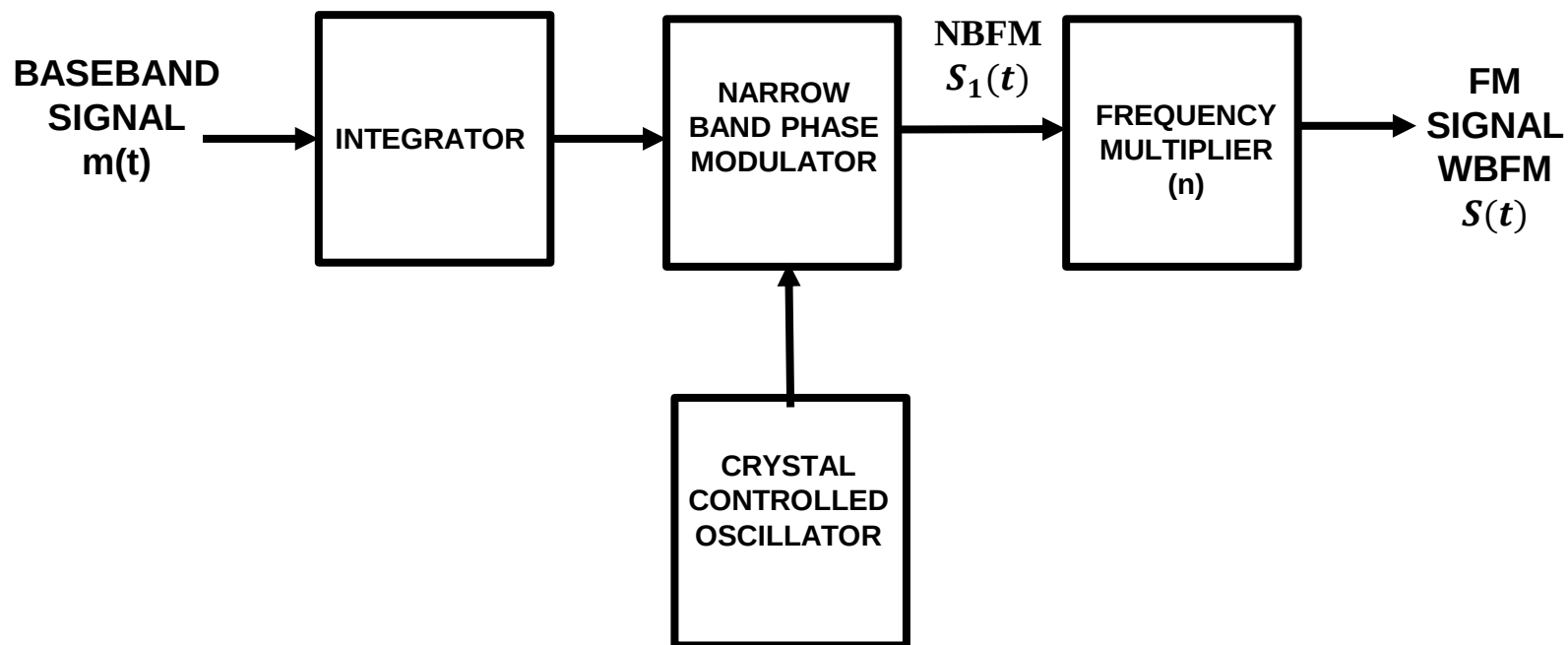


Fig.5.4.6 [Indirect method of WBFM generation]

Armstrong Method of FM Generation contd.

- In the block diagram of an Indirect FM generation, the base-band signal $m(t)$ is first integrated and then used to phase-modulate a crystal controlled oscillator.
- To minimize the distortion inherent in the phase-modulator, the maximum phase deviation/modulation index β is kept small ($\beta \leq 0.3$), thereby resulting a NBFM-wave.
- This signal is next multiplied in frequency by means of a frequency multiplier, to produce the desired WBFM-wave.



Armstrong Method of FM Generation contd.

- Let $S_1(t)$ denote the output of phase modulator,

$$S_1(t) = A_1 \cos \left[2\pi f_1 t + 2\pi K_f \int_0^t m(t) dt \right] \quad [\text{Eq.5.4.16}]$$

Where, $f_1 \rightarrow$ frequency of the crystal oscillator

$K_f \rightarrow$ modulation sensitivity of FM-modulator

$\beta_1 = \frac{\Delta f}{f_m} \rightarrow$ modulation index (≤ 0.3)

- The phase modulator output is next multiplied n -times in frequency by the *Frequency-Multiplier*, producing the desired WBFM-wave.



Armstrong Method of FM Generation contd.

$$\begin{aligned}\therefore S(t) &= A_C \cos \left[2\pi(nf_1) + 2\pi nK_f \int_0^t m(t) \cdot dt \right] \\ &= A_C \cos[2\pi(nf_1) + n\beta_1 \sin(2\pi f_m t)] \\ &= A_C \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]\end{aligned}\quad [\text{Eq.5.4.17}]$$

Where, $\beta = n\beta_1$ and $nf_1 = f_c$

- Thus by choosing ' n ' properly, we may set the final value of the modulation index β at any desired value.



FM Bandwidth

□ Carson's Rule For FM Bandwidth:

- *Carson's rule of bandwidth states that to transmit 98% of total information contained in an FM signal, the bandwidth required is finite.*
- *Carson's rule approximates the bandwidth necessary to transmit FM wave as twice the sum of the peak frequency deviation and the maximum modulating-signal frequency.*

Mathematically, $B_T = (2\Delta f + 2f_m)$ [Eq.5.4.18]

- For NBFM i.e. for small values of Δf , $B_T \simeq 2f_m$
- For WBFM i.e. for large values of Δf , $B_T \simeq 2\Delta f$



FM Bandwidth contd.

- We can express the transmission bandwidth as,

$$\begin{aligned} B_T &= (2\Delta f + 2f_m) \\ &= 2f_m \left(\frac{\Delta f}{f_m} + 1 \right) \end{aligned}$$

$$\Rightarrow B_T = 2f_m(\beta + 1) \quad [\text{Eq.5.4.19}]$$

- Thus in an FM signal, the transmission bandwidth (B_T) depends on the modulation index (β).



Comparison of AM, FM and PM

Sl. No.	AM	FM & PM
1.	Amplitude and transmitted power of AM signal change with modulation.	Amplitude and transmitted power of FM or PM signal do not change with modulation.
2.	AM signal has an envelope that is used to demodulate the modulating signal	FM or PM signal does not have an envelope, and the demodulator need not to respond to amplitude variations.
3.	AM signals are more susceptible to noise, because information is stored as amplitude variations rather than frequency variations.	FM or PM signals are less susceptible to noise, because information is stored as frequency/phase variations rather than amplitude variations.



Comparison of AM, FM and PM contd.

Sl. No.	AM	FM & PM
1.	Modulation index can not be changed automatically.	The modulation index can be varied to obtain greater SNR.
2.	AM signals occupy lesser bandwidth.	FM or PM signals occupy more bandwidth.
3.	Modulation can be accomplished at low power or high power levels.	Modulation can be accomplished at low power levels.



Applications of FM and PM

- FM is widely used for radio broadcasting, sound signal transmission in TV broadcast, two-way fixed and mobile radio systems, analog cellular communications systems and satellite communications.
- Phase modulation is mostly used in data communications and generating FM in FM-transmitters.

